



Computer Science Department
COMP233-Discrete Mathematics/ Spring 2026
Section #4 - Mid #1 - 7 May: Chapters 2, 3, 4
Time: 60 Minutes | Total Marks: 57

Student Name:

Student ID:

Key Answer

Question 1#: Circle **ALL** correct answers for each branch. Fill your final answers in the table in the answer sheet. **[15 Marks]**

Question	1	2	3	4	5	6	7	8	9	10
Answer	b	b, d	b, c	c	b	c	c	a, c	b, c, d	a

Question #2: Basic Logic Section

[15 Marks]

A) Formal Negation (6 Marks):

1. $\exists$ computer c such that c does not have a CPU. (3 Marks)
2. $\forall x \in \mathbb{R}, x^2 \neq 2$. (3 Marks)

B) Logical Equivalence Reasons (9 Marks):

Step 1: $(\sim(\sim p) \vee \sim q) \wedge (p \vee q) \rightarrow$ De Morgan's laws

Step 2: $(p \vee \sim q) \wedge (p \vee q) \rightarrow$ Double negative law (2 Marks)

Step 3: $p \vee (\sim q \wedge q) \rightarrow$ Distributive law (2 Marks)

Step 4: $p \vee (q \wedge \sim q) \rightarrow$ Commutative law for $\wedge$ (2 Marks)

Step 5: $p \vee c \rightarrow$ Negation law (2 Marks)

Step 6: $p \rightarrow$ Identity law (1 Marks)

Question #3: Concepts Section

[10 Marks]

A) Statement Variations (4 Marks):

1. **Contrapositive:** If n is not odd, then n is not a prime number or n is not greater than 2. (2 Marks) // If n is even, then n is not prime or $n \leq 2$.
2. **Negation:** n is a prime number and $n > 2$, and n is not odd. (2 Marks) // n is a prime number and $n > 2$, and n is even.

B) Truth Table & Validity (6 Marks):

p	q	$p \rightarrow q$	$\sim p$	$\sim q$
T	T	T	F	F
T	F	F	F	T
F	T	T	T	F
F	F	T	T	T

- **Result:** The argument is Invalid (Inverse Error). (2 Marks)
- **Justification:** In the 3rd row, both premises $(p \rightarrow q)$ and $(\sim p)$ are **True**, but the conclusion $(\sim q)$ is **False**. (2 Marks)



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Question #4: Proofs Section

[17 Marks]

A) True/False (4 Marks):

3. **True:** For any integer x , let $y = -x$. Since $x + (-x) = 0$ and $-x$ is an integer, the statement holds. (2 Marks)
4. **False:** There is no single integer y that works for every x . (If x changes, y must change to keep the sum 0). (2 Marks)

B.1) Divisibility Proof (6 Marks):

Suppose $n = 4k + 1$ for some integer k .

Then $n^2 - 1$

$$= (4k + 1)^2 - 1$$

$$= (16k^2 + 8k + 1) - 1$$

$$= 16k^2 + 8k$$

$$= 8(2k^2 + k)$$

Since k is an integer, $(2k^2 + k)$ is an integer (let it be m).

Thus,

$$n^2 - 1 = 8m, \text{ which means } \mathbf{8 \text{ divides } (n^2 - 1)}.$$



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B.2) Consecutive Integers Proof (7 Marks):

- **Let** the two consecutive integers be n and $n+1$.
- By the **Quotient-Remainder Theorem** with $d=3$, any integer n is either $3q$, $3q+2$ or $3q+2$ for some integer q .
 - **Case 1 ($n = 3q$):** The product $n(n+1)$
$$= 3q(3q+1)$$
$$= 3[q(3q+1)].$$

This is in the form $3k$ (where $k = q(3q+1)$ is an integer).
 - **Case 2 ($n = 3q+1$):** The product $n(n+1) = (3q+1)(3q+1+1)$
$$= (3q+1)(3q+2)$$
$$= 9q^2 + 6q + 3q + 2$$
$$= 9q^2 + 9q + 2$$
$$= 3[3q^2 + 3q] + 2.$$

This is in the form $3k + 2$ (where $k = 3q^2 + 3q$ is an integer).
 - **Case 3 ($n = 3q+2$):** The product $n(n+1)$
$$= (3q+2)(3q+2+1)$$
$$= (3q+2)(3q+3)$$
$$= (3q+2) \cdot 3(q+1)$$
$$= 3[(3q+2)(q+1)].$$

This is in the form $3k$ (where $k = (3q+2)(q+1)$ is an integer).
- In all cases, the product $n(n+1)$ has the form $3k$ or $3k+2$.